

LEADFLOW AI

Five-Day AI OS Sprint

Final System, Evidence, Demo & Submission Guide

Awais Saeed | Applied AI Engineer Candidate

Target user: Non-developer Inbound SDR

Secondary user: RevOps Manager

Final state	What is evidenced
Day 1	Problem discovery, workflow, baseline, 12-case benchmark, scope and non-goals
Day 2	Architecture, contracts, deterministic scoring, bounded LLM, approval gate, audit trail
Day 3	Real DNS + website integrations, end-to-end UI, benchmark validation, reproducibility
Day 4	Failure injection, root-cause analysis, hardening, before/after regression, metrics
Day 5	Handoff, reproducibility, README/runbook, case study, demo plan, 2-week adoption/quality plan

Purpose: a reviewer can understand, run, trust, evaluate, and improve the system without the candidate beside them.

1. Executive Summary

LeadFlow AI is an auditable, human-governed AI operating system for inbound B2B sales lead triage. It addresses the recurring interval between inbound lead submission and an approved first-touch sales action.

The system combines deterministic business logic with bounded generative AI. Untrusted lead text and external website content are isolated from the trusted qualification logic. A human approval gate remains mandatory before CRM dispatch authorization.

Evidence	Current recorded result
Day 1 manual baseline	16m45s per lead; 7 browser tabs; 5 context switches
Day 1 simple ChatGPT baseline	11m10s per lead; 5.7/10 quality
Final automated tests	56/56 passed
Final benchmark	12/12 cases matched tier, score and route
Day 4 break suite	10/10 deliberate failure scenarios handled as designed
Measured benchmark processing time	2.3s average across 11 processable cases
Measured comparison	~98% reduction versus 16m45s manual baseline
Human approval	PENDING_REVIEW / PREVIEW_ONLY before approval; APPROVED / APPROVED_FOR_DISPATCH after approval

Important: benchmark and performance numbers are assessment-environment measurements. They are not production adoption or revenue outcomes.

2. Day 1 - Discover, Map, Baseline

What was built

- Defined the recurring bottleneck: the manual qualification and enrichment interval between inbound lead submission and approved first-touch action.
- Mapped trigger, input, judgment, tool, approval, output, and exception paths.
- Established a controlled baseline and a 12-case benchmark.
- Defined the deterministic ICP scoring formula and explicit tier/routing rules.
- Defined success metrics and non-goals for a five-day scope.

Current workflow

Inbound lead -> read raw input -> verify identity/domain -> research account -> qualify -> duplicate/CRM checks -> prepare first touch -> review -> route/dispatch -> log

Proposed workflow

Inbound lead -> validate -> risk check -> external verification -> synthetic enrichment -> deterministic ICP -> bounded draft -> claim validation -> human approval -> CRM-ready payload -> audit

Baseline

Method	Observed baseline
Manual SDR workflow	16m45s average per lead; 8.0/10 quality; 7 browser tabs; 5 context switches
Simple ChatGPT-assisted workflow	11m10s average per lead; 5.7/10 quality

Benchmark discipline

The benchmark includes normal, edge, and adversarial cases, including enterprise, SMB exception, free-mail verification, sparse data, German/GDPR, competitor, prompt injection, academic, and malformed input scenarios.

3. Day 2 - Design the System and Ship v0

Architecture

Raw Lead

- > Pydantic Validation
- > Risk Detector
- > Synthetic Enrichment
- > Deterministic ICP Scorer
- > Bounded LLM / Deterministic Fallback
- > Claim Validator
- > Human Approval Gate
- > CRM-ready Payload
- > Append-only SQLite Audit Log

Key engineering decisions

Decision	Why
Pydantic v2	Typed boundaries and clear validation errors
Deterministic ICP scoring	Business qualification cannot be changed by LLM output
Bounded LLM	LLM drafts only; verified context is supplied and commercial claims are validated after generation
Deterministic fallback	Core workflow remains usable when the LLM provider is unavailable
Human approval	No CRM dispatch authorization without explicit human action
SQLite audit trail	Simple local append-only evidence store for v0
Synthetic enrichment	Safe assessment data boundary; no real customer PII
Real DNS/website checks later added in Day 3	Adds independently observable external signals without requiring a proprietary enrichment account

Critical governance rule

Before approval: PENDING_REVIEW + PREVIEW_ONLY + dispatch_authorized=false

After approval: APPROVED + APPROVED_FOR_DISPATCH + dispatch_authorized=true

4. Day 3 - Build the Working Core

Two real external integrations

- Real DNS-over-HTTPS verification using Cloudflare public DNS.
- Real HTTP website metadata retrieval with bounded timeout and redirect handling.

External-data trust boundary

External DNS and website results are treated as untrusted evidence. They cannot modify scoring, routing, approval, system instructions, or CRM authorization. Website text is never trusted as an executable instruction.

Non-developer interface

- Lead input form
- Verification and enrichment display
- ICP score and routing
- Draft and claim status
- Human approval/edit/quarantine actions
- CRM preview/authorization status
- Audit trail summary

Final Day 3 proof

- 12/12 benchmark cases matched expected tier, score and route after evaluator repair.
- 41/41 tests were the Day 3 final suite before Day 4 expansion.
- A proxy-user flow demonstrated the full processing and approval path.

5. Day 4 - Evaluate, Break, and Harden

Day 4 deliberately attacked the system rather than relying on the happy path. The goal was to explain what fails, why it fails, and what changed after hardening.

Failure suite

Failure	Expected containment / response
DNS failure	Bounded external failure; continue where safe; log and show verification unavailable
Website timeout	Bounded timeout; no hang; structured timeout message; continue safely
LLM unavailable	Deterministic fallback; transparent source label; claim validation still runs
Prompt injection	Treat as untrusted; quarantine where required; no rule change
Malicious website content	External content cannot alter deterministic scoring or approval
Unsafe commercial claim	Claim validator blocks approval until corrected
Malformed input	Friendly validation error; no uncontrolled 500
Enrichment miss	No fabricated facts; explicit unavailable/verification state
Quarantine bypass	Approval remains blocked
Unsafe edited draft	Final body revalidated before approval

Final Day 4 evidence

Metric	Final evidence
Automated suite	56/56 passed
12-case benchmark	12/12 passed
Deliberate break suite	10/10 passed
Claim safety metric	9/9 generated drafts passed; malformed TC-12 excluded from denominator

Measured benchmark latency	2.3s average across 11 processable cases
Website timeout	5-second configured bound in final documentation
Approval gate	Verified pre-approval unauthorized and post-approval authorized

Before/after learning loop

Baseline -> break the system -> observe failure -> identify root cause -> implement targeted hardening -> rerun same case -> record regression result

6. Day 5 - Handoff, Prove Value, and Present

Day 5 packages the system for someone who did not build it. The goal is operational handoff, not more feature work.

Final repository map

```
leadflow-ai-final/
├── README.md
├── QUICKSTART.md
├── start_leadflow.bat
├── day1/
├── day2/
├── day3/
├── day4/
├── day5/
├── docs/
├── operations/
├── demo/
├── case_study/
└── evidence/
```

Three-step setup

1. Open a terminal in the repository root and install dependencies: `cd day2` then `pip install -r requirements.txt`
2. Start the app: double-click `start_leadflow.bat`, or run `python -m uvicorn backend.app.main:app --host 0.0.0.0 --port 8000 --reload` from `day2`.
3. Open `http://localhost:8000` and run TC-01 or a normal lead through review and approval.

First-run operator workflow

4. Select a benchmark preset or enter a lead manually.
5. Click Process Inbound Lead.
6. Review DNS, website, enrichment, ICP, routing, and draft.
7. Check claim status and risk status.
8. Before approval confirm `PENDING_REVIEW` and `PREVIEW_ONLY`.
9. Approve, edit, or quarantine according to the system result.
10. After approval confirm `APPROVED_FOR_DISPATCH` and review the audit record.

7. How to Run the Final System

Windows - recommended

1. Extract the final ZIP.
2. Open Command Prompt/PowerShell in the extracted folder.
3. Run: `.\start_leadflow.bat`
4. Open: `http://localhost:8000`

If the startup batch file is not used:

```
cd day2
pip install -r requirements.txt
python -m uvicorn backend.app.main:app --host 0.0.0.0 --port 8000 --reload
```

Verify health

`http://localhost:8000/health`

Run the full test suite

```
cd day2
pytest -q
```

Run the 12-case benchmark

```
python day3/benchmark/run_12_case_benchmark.py
```

Run Day 4 break tests

```
python day4/break_tests/run_break_tests.py
```

8. Five-Minute Screen Recording - Exact Plan

The repository contains the demo script/checklist/results, but the five-minute video must be recorded separately if the assessment platform requires an actual video file. Do not claim a video is included unless you actually record it.

Recommended recording setup

- Use OBS Studio, Windows Game Bar, or another screen recorder available on your machine.
- Record the application browser window and microphone narration.
- Close personal tabs, bookmarks, notifications, and unrelated applications.
- Do not display API keys, `.env` contents, terminal secrets, or personal filesystem paths.
- Use a clean benchmark lead such as TC-01 and prepare one short failure scenario.

Five-minute timeline

Time	Show / say
0:00-0:30	Problem: SDR spends time moving between research tabs before an approved first touch. Mention the Day 1 baseline.
0:30-1:00	Architecture: validation -> risk -> DNS -> website -> enrichment -> deterministic ICP -> draft -> claims -> human approval -> CRM payload -> audit.
1:00-2:00	Live processing: choose TC-01, click Process Inbound

	Lead, show real DNS/website results, enrichment, score, tier and route.
2:00-2:45	Show draft, source/fallback status, and claim validation. Explain that LLM output cannot change deterministic scoring.
2:45-3:30	Show PENDING_REVIEW / PREVIEW_ONLY. Explain human approval is mandatory. Click Approve and show APPROVED_FOR_DISPATCH.
3:30-4:15	Show audit event history and CRM-ready payload. Explain that v0 does not connect to a live CRM.
4:15-4:45	Break demo: trigger website timeout, LLM fallback, or unsafe claim. Show controlled failure/fallback and message.
4:45-5:00	Results: 56/56 tests, 12/12 benchmark, 10/10 break suite. State limitations and next iteration.

Narration example

“LeadFlow AI addresses a recurring inbound sales bottleneck. Instead of asking an SDR to manually research, qualify, draft, and prepare a lead, the system moves the repetitive work into a governed pipeline. Deterministic business rules remain authoritative, external data is treated as untrusted, the LLM is bounded to drafting, and a human must approve before CRM dispatch is authorized.”

What to demonstrate visually

- Lead form / TC-01 preset
- DNS VERIFIED
- Website metadata result
- Synthetic enrichment source
- ICP score + tier + route
- Draft + fallback/LLM indicator
- Claim validation
- PENDING_REVIEW / PREVIEW_ONLY
- Approve & Dispatch
- APPROVED / APPROVED_FOR_DISPATCH
- Audit trail
- One failure or fallback

9. How to Record and Submit the Video

Option A - OBS Studio

11. Install/open OBS Studio.
12. Create a Scene named LeadFlow Demo.
13. Add a Window Capture or Display Capture for the browser/application.
14. Add your microphone as an Audio Input Capture.
15. Open the LeadFlow UI before recording.
16. Start Recording.
17. Follow the 5-minute script exactly once without stopping if possible.
18. Stop Recording and review the file for readable UI, clear voice, and no sensitive information.

19. Export/keep MP4 or another assessment-accepted format.

Option B - Windows Game Bar

20. Open the LeadFlow browser window.
21. Press Win + G.
22. Enable microphone if narration is required.
23. Start recording and follow the same five-minute script.
24. Stop, review, and save the clip.

Video filename

LeadFlow_AI_5_Minute_Demo_Awais_Saeed.mp4

Before uploading the video

Check	Pass condition
Length	Approximately 5 minutes
Audio	Voice is understandable
UI	Readable at normal zoom
Secrets	No keys/tokens/passwords visible
Personal data	No real customer PII shown
Paths	No personal Windows path visible
Flow	Happy path + one failure/fallback
Governance	Pre-approval locked; post-approval authorized
Results	Numbers match final evidence package

10. Final Submission Checklist

Use this checklist immediately before submission.

- ☐ Final repository contains Days 1–5 and the actual source used for the final tests.
- ☐ README and QUICKSTART work from a fresh extraction directory.
- ☐ No personal filesystem paths appear in user-facing documentation or evidence.
- ☐ No secrets, API keys, tokens, .env files with credentials, or private data are included.
- ☐ Run `pytest -q` and record the actual final result.
- ☐ Run the 12-case benchmark and keep the machine-generated results.
- ☐ Run the Day 4 break suite and keep the machine-generated results.
- ☐ Confirm pre-approval is `PENDING_REVIEW + PREVIEW_ONLY + dispatch_authorized=false`.
- ☐ Confirm post-approval is `APPROVED + APPROVED_FOR_DISPATCH + dispatch_authorized=true`.
- ☐ Confirm the case study uses measured results as measured results and targets as targets.
- ☐ Confirm limitations are explicit: synthetic data, no live CRM, assessment environment, human approval required.
- ☐ Record the five-minute screen demo separately if required.
- ☐ Review the full video before uploading.
- ☐ Zip only the clean final repository and required evidence.

11. How the Five-Day Work Maps to the Evaluation Rubric

Rubric	Where to point the evaluator
Problem Leverage and Scope - 15%	Day 1 problem definition, workflow map,

	baseline_measurements.csv, test_cases.json, explicit non-goals
System Architecture and Engineering - 20%	Day 2 architecture.md, architecture_decisions.md, data contracts, tool boundaries, security model
Working Product and Reliability - 20%	Day 3 real integrations, Day 4 hardening, 56 tests, logs, fallbacks, approval gate, clean-room handoff
Evaluation and Learning - 20%	Day 1 baseline + 12 cases, Day 4 break suite, root-cause analysis, metrics, before/after regression
Non-Developer UX and Adoption - 15%	Web UI, docs/user_readme.md, QUICKSTART, operator_runbook, proxy-user execution/feedback
Ownership and Communication - 10%	AI collaboration notes, architecture decisions, case study, demo script, limitations, improvement plan

12. What You Should Say Honestly in the Presentation

- This is a controlled five-day assessment system, not a production deployment.
- The internal enrichment source is synthetic.
- The CRM integration is preview-only; there is no live Salesforce/HubSpot mutation.
- External DNS and website checks are network dependent.
- The prompt-injection detector is deterministic and therefore not a complete semantic security solution.
- Human approval remains mandatory by design.
- The two-week adoption and quality metrics are future operating targets, not achieved post-deployment outcomes.

13. Final Presentation Closing

Recommended closing statement:

“I started with a recurring operational bottleneck, measured a baseline, built a governed end-to-end workflow, connected real external verification, deliberately broke the system, hardened the failure paths, and validated the result with a benchmark and regression suite. The system is intentionally constrained: it uses synthetic enrichment, produces CRM-ready output instead of mutating a live CRM, and keeps a human in control of final approval. The next step would be controlled production integration with monitoring and broader evaluation.”